Where Bayesian Changed My Decision:

The Situation

In Part 1 of this assignment, I was asked to estimate the churn rate for Group A_small, a sample of only 40 Month-to-month customers. The VP wanted to know: should we invest in a targeted retention campaign for this segment?

Using MLE, the answer was simple: 15 out of 40 customers churned, so the estimated churn rate is $\theta = 0.375$, or 37.5%. Based on this alone, I would have confidently recommended the campaign, a 37.5% churn rate is very high and seems to clearly justify intervention.

What the Bayesian Answer Showed

When I computed the full posterior using a Beta(2,8) prior (encoding a prior belief that churn rates are generally around 20%), the posterior was Beta(17, 33), which gave a mean of about 0.34 and a 94% HDI of roughly [0.22, 0.47].

That interval stretching from 22% to 47% changed my thinking entirely. The MLE gave me 37.5% with no uncertainty attached. The Bayesian posterior told me I was actually quite uncertain: the true rate could plausibly be 22% (moderate, perhaps not worth a costly campaign) or 47% (severe, definitely worth acting on). With only 40 observations, I simply did not have enough data to tell the difference.

The Decision Change

Based on MLE alone: I would have initiated the campaign immediately, treating 37.5% as a known fact.

Based on the Bayesian posterior: I would instead recommend collecting more data first. Specifically, waiting until we have at least 200 - 300 observations from this segment, before committing campaign budget. The HDI is too wide to support a confident financial decision.

The Mechanism

The mechanism is that MLE silently discards sample size information. It produces a point estimate that looks equally confident whether it came from 40 observations or 4000. The Bayesian posterior, in contrast, encodes sample size directly in the parameters of the Beta distribution: $\alpha + \beta$ = prior_counts + n. Small n keeps the posterior wide, large n concentrates it. This means the posterior automatically communicates how much we should trust the estimate, something the MLE simply cannot do on its own.

This is not just a philosophical difference. In a business context, the width of the posterior is actionable information. It tells me whether I have enough evidence to make a decision now, or whether I need to gather more data first. The MLE gives me a number; the Bayesian posterior tells me whether that number is reliable enough to act on.